

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
nmmayorov@nmmayorov:~$ ls /etc/ > file.txt
nmmayorov@nmmayorov:~$ ls >> file.txt
nmmayorov@nmmayorov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
```

Рисунок 1: Запись в файл

```
nmmayorov@nmmayorov:~$ grep .conf file.txt > conf.txt
nmmayorov@nmmayorov:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
```

Рисунок 2: Поиск расширения .conf


```
/home/nmmayorov/work/study/2025-2026/Операционные системы/2026-1--stud
y--os-intro/project-personal/stage03/report/_resources/csl
/home/nmmayorov/work/study/2025-2026/Операционные системы/2026-1--stud
y--os-intro/project-personal/stage03/report/bib/cite.bib
/home/nmmayorov/work/study/2025-2026/Операционные системы/2026-1--stud
y--os-intro/project-personal/stage04/report/_resources/csl
/home/nmmayorov/work/study/2025-2026/Операционные системы/2026-1--stud
y--os-intro/project-personal/stage04/report/bib/cite.bib
/home/nmmayorov/work/study/2025-2026/Операционные системы/2026-1--stud
y--os-intro/project-personal/stage05/report/_resources/csl
/home/nmmayorov/work/study/2025-2026/Операционные системы/2026-1--stud
y--os-intro/project-personal/stage05/report/bib/cite.bib
/home/nmmayorov/work/study/2025-2026/Операционные системы/2026-1--stud
y--os-intro/project-personal/stage06/report/_resources/csl
/home/nmmayorov/work/study/2025-2026/Операционные системы/2026-1--stud
y--os-intro/project-personal/stage06/report/bib/cite.bib
/home/nmmayorov/work/study/2025-2026/Операционные системы/2026-1--stud
y--os-intro/presentation/report/_resources/csl
/home/nmmayorov/work/study/2025-2026/Операционные системы/2026-1--stud
y--os-intro/presentation/report/bib/cite.bib
/home/nmmayorov/git-extended/.git/hooks/commit-msg.sample
/home/nmmayorov/git-extended/.git/config
/home/nmmayorov/conf.txt
nmmayorov@nmmayorov:~$
```

Рисунок 3: Поиск файлов

```
find: '/etc/grub.d': Permission denied
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hpclip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
nmmayorov@nmmayorov:~$  
nmmayorov@nmmayorov:~$ find ~ -name "log*" > logfile &  
[1] 6310  
nmmayorov@nmmayorov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
nmmayorov@nmmayorov:~$ rm logfile  
nmmayorov@nmmayorov:~$
```

Рисунок 5: Фоновый запуск процесса

```
nmmayorov@nmmayorov:~$  
nmmayorov@nmmayorov:~$ find ~ -name "log*" > logfile &  
[1] 6310  
nmmayorov@nmmayorov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
nmmayorov@nmmayorov:~$ rm logfile  
nmmayorov@nmmayorov:~$  
nmmayorov@nmmayorov:~$ gedit &  
[1] 6342  
nmmayorov@nmmayorov:~$ ps | grep gedit  
    6342 pts/1    00:00:00 gedit  
nmmayorov@nmmayorov:~$ kill 6342  
[1]+  Terminated          gedit  
nmmayorov@nmmayorov:~$  
nmmayorov@nmmayorov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays
    the amount of space available on the file system containing
    each file name argument.  If no file name is given, the space
    available on all currently mounted file systems is shown.
    Space is shown in 1K blocks by default, unless the environment
    variable POSIXLY_CORRECT is set, in which case 512-byte blocks
    are used.

    If an argument is the absolute file name of a device node con-
    taining a mounted file system, df shows the space available on
    that file system rather than on the file system containing the
    device node.  This version of df cannot show the space avail-
    able on unmounted file systems, because on most kinds of sys-
    tems doing so requires non-portable intimate knowledge of file
    system structures.

OPTIONS
    Show information about the file system on which each FILE re-
    sides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short
    options too.

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILEs, recursively for
    directories.

    Mandatory arguments to long options are mandatory for short
    options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

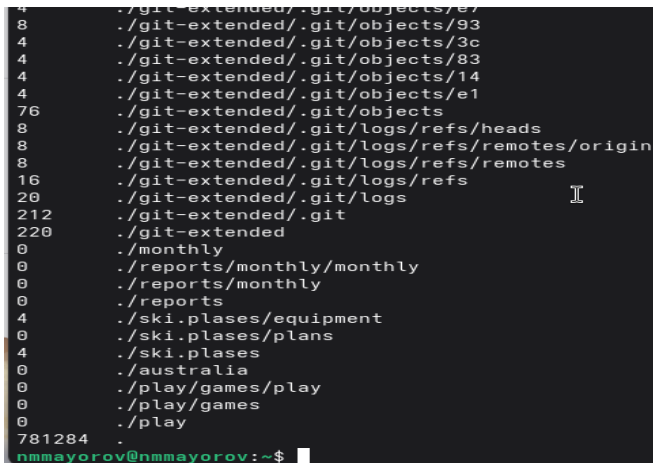
    --apparent-size
        print apparent sizes rather than device usage; although
        the apparent size is usually smaller, it may be larger
        due to holes in ('sparse') files, internal fragmenta-
        tion, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM'
        prints sizes in units of 1,048,576 bytes; see SIZE for-
    Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
nmmayorov@nmmayorov:~$ df
nmmayorov@nmmayorov:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 91387280 63208464 60% /
devtmpfs        4011304      0    4011304  0% /dev
tmpfs           4035676      92    4035584  1% /dev/shm
tmpfs           1614272     1952    1612320  1% /run
tmpfs           1024         0      1024     0% /run/credentials/syst
emd-journald.service
/dev/nvme0n1p3 155186176 91387280 63208464 60% /home
tmpfs           4035680      80    4035600  1% /tmp
/dev/nvme0n1p2 1992552     518800    1352512 28% /boot
/dev/loop2      49280       49280         0 100% /var/lib/snapd/snap/s
napd/25935
/dev/loop0      115712     115712         0 100% /var/lib/snapd/snap/h
ugo/25676
/dev/loop1      68480       68480         0 100% /var/lib/snapd/snap/c
ore24/1349
tmpfs           1024         0      1024     0% /run/credentials/syst
emd-resolved.service
tmpfs           807132      172    806960  1% /run/user/1066
/dev/sr0        2677920     2677920         0 100% /run/media/nmmayorov/
Fedora-WS-Live-43
nmmayorov@nmmayorov:~$
```

Рисунок 9: Справка по команде du



```
4      ./git-extended/.git/objects/e7
8      ./git-extended/.git/objects/93
4      ./git-extended/.git/objects/3c
4      ./git-extended/.git/objects/83
4      ./git-extended/.git/objects/14
4      ./git-extended/.git/objects/e1
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
781284 .
nmmayorov@nmmayorov:~$
```

Рисунок 10: Запуск команды du


```
/home/nmmayorov/git-extended/.git/objects/0d
/home/nmmayorov/git-extended/.git/objects/2d
/home/nmmayorov/git-extended/.git/objects/35
/home/nmmayorov/git-extended/.git/objects/18
/home/nmmayorov/git-extended/.git/objects/fd
/home/nmmayorov/git-extended/.git/objects/e7
/home/nmmayorov/git-extended/.git/objects/93
/home/nmmayorov/git-extended/.git/objects/3c
/home/nmmayorov/git-extended/.git/objects/83
/home/nmmayorov/git-extended/.git/objects/14
/home/nmmayorov/git-extended/.git/objects/e1
/home/nmmayorov/git-extended/.git/logs
/home/nmmayorov/git-extended/.git/logs/refs
/home/nmmayorov/git-extended/.git/logs/refs/heads
/home/nmmayorov/git-extended/.git/logs/refs/remotes
/home/nmmayorov/git-extended/.git/logs/refs/remotes/origin
/home/nmmayorov/monthly
/home/nmmayorov/reports
/home/nmmayorov/reports/monthly
/home/nmmayorov/reports/monthly/monthly
/home/nmmayorov/ski.places
/home/nmmayorov/ski.places/equipment
/home/nmmayorov/ski.places/plans
/home/nmmayorov/australia
/home/nmmayorov/play
/home/nmmayorov/play/games
/home/nmmayorov/play/games/play
nmmayorov@nmmayorov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.